Class 12 Mathematics

Chapter 11: Three Dimensional Geometry

Practice Set A: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE 2024 — 1 mark

Direction ratios of a vector parallel to the line $\frac{x-1}{2} = -y = \frac{2z+1}{6}$ are:

(A) $2, -1, 6$ **(B)** $2, 1, 6$

(C) $2, 1, 3$ **(D)** $2, -1, 3$

Formula / idea: In $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ the coefficients of x, y, z are 1; DRs are a, b, c.

1 Write $-y$ as $\frac{y}{-1}$.

$$-y = \frac{y-0}{-1}$$

2 Divide top and bottom of the last fraction by 2.

$$\frac{2z+1}{6} = \frac{z+\frac{1}{2}}{3}$$

3 Write the line in standard form.

$$\frac{x-1}{2} = \frac{y-0}{-1} = \frac{z+\frac{1}{2}}{3}$$

4 Read the denominators as direction ratios. ✓ 1

$$\text{DRs} = 2,\ -1,\ 3$$

5 Hence the answer is option (D).

Answer **(D)** $2, -1, 3$

Watch out: Reading 6 from $\frac{2z+1}{6}$ (option A) without first making the coefficient of z equal to 1.

2 MCQ CBSE 2024 1 mark

If a line makes an angle of 30° with the positive direction of x-axis, 120° with the positive direction of y-axis, then the angle which it makes with the positive direction of z-axis is:

(A) 90° **(B)** 120°

(C) 60° **(D)** 0°

Formula / idea: If a line makes angles α, β, γ with the axes, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

1 Use the identity for direction cosines.
$\cos^2 30^\circ + \cos^2 120^\circ + \cos^2\gamma = 1$

2 Put the values of the cosines.
$\left(\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + \cos^2\gamma = 1$

3 Simplify.
$\frac{3}{4} + \frac{1}{4} + \cos^2\gamma = 1 \Rightarrow \cos^2\gamma = 0$

4 Solve for γ. ✓ 1
$\cos\gamma = 0 \Rightarrow \gamma = 90^\circ$

5 Hence the answer is option (A).

Answer **(A)** 90°

Watch out: Picking 0° (option D) by confusing $\cos\gamma = 0$ with $\gamma = 0$.

3 MCQ CBSE 2024 1 mark

The Cartesian equation of a line passing through the point with position vector $\vec{a} = \hat{i} - \hat{j}$ and parallel to the line $\vec{r} = \hat{i} + \hat{k} + \mu(2\hat{i} - \hat{j})$ is:

(A) $\frac{x-2}{1} = \frac{y+1}{0} = \frac{z}{1}$

(B) $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{0}$

(C) $\frac{x+1}{2} = \frac{y+1}{-1} = \frac{z}{0}$

(D) $\frac{x-1}{2} = \frac{y}{-1} = \frac{z-1}{0}$

Formula / idea: Line through (x_1, y_1, z_1) with DRs a, b, c: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$.

1 Read the point from the position vector.
$\vec{a} = \hat{i} - \hat{j} \Rightarrow (1, -1, 0)$

2 A parallel line has the same direction vector.
$\vec{b} = 2\hat{i} - \hat{j} + 0\hat{k} \Rightarrow \text{DRs } 2, -1, 0$

3 Write the Cartesian form. ✓ 1
$\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-0}{0}$

4 Hence the answer is option (B).
A zero denominator only means z stays 0 on the line.

Answer **(B)** $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{0}$

Watch out: Using the point $(1, 0, 1)$ of the given line (option D) instead of the given point $(1, -1, 0)$.

4 MCQ CBSE 2023 1 mark

The angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$ is:

(A) $0°$ **(B)** $30°$
(C) $45°$ **(D)** $90°$

Formula / idea: Lines with DRs a_1, b_1, c_1 and a_2, b_2, c_2 are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1 Divide $2x = 3y = -z$ by 6 to get DRs.
$\frac{x}{3} = \frac{y}{2} = \frac{z}{-6} \Rightarrow 3, 2, -6$

2 Divide $6x = -y = -4z$ by 12 to get DRs.
$\frac{x}{2} = \frac{y}{-12} = \frac{z}{-3} \Rightarrow 2, -12, -3$

3 Find $a_1a_2 + b_1b_2 + c_1c_2$. ✓ 1
$3(2) + 2(-12) + (-6)(-3) = 6 - 24 + 18 = 0$

4 So $\cos\theta = 0$, the lines are perpendicular. Answer: option (D).

Answer **(D)** $90°$

Watch out: Taking DRs $2, 3, -1$ straight from $2x = 3y = -z$; the DRs are the reciprocals $\frac{1}{2}, \frac{1}{3}, -1$.

5 MCQ CBSE 2024 1 mark

The lines $\frac{1-x}{2}=\frac{y-1}{3}=\frac{z}{1}$ and $\frac{2x-3}{2p}=\frac{y}{-1}=\frac{z-4}{7}$ are perpendicular to each other for p equal to:

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) 2

(D) 3

Formula / idea: Perpendicular lines: $a_1a_2+b_1b_2+c_1c_2=0$, with both lines first in standard form.

1 Make the coefficient of x equal to $+1$ in the first line.
$\frac{x-1}{-2}=\frac{y-1}{3}=\frac{z}{1}\Rightarrow \text{DRs } -2,3,1$

2 Divide top and bottom by 2 in the second line.
$\frac{x-\frac{3}{2}}{p}=\frac{y}{-1}=\frac{z-4}{7}\Rightarrow \text{DRs } p,-1,7$

3 Apply the perpendicular condition. ✓ 1
$-2p-3+7=0\Rightarrow p=2$

4 Hence the answer is option (C).

Answer **(C)** 2

Watch out: Keeping $2p$ as a DR without dividing $\frac{2x-3}{2p}$ by 2; that gives $p=1$, which is not an option.

6 Assertion-Reason CBSE 2023 1 mark

Assertion (A): If a line makes angles α,β,γ with the positive directions of the coordinate axes, then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$. **Reason (R):** The sum of squares of the direction cosines of a line is 1.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Direction cosines $l=\cos\alpha, m=\cos\beta, n=\cos\gamma$ satisfy $l^2+m^2+n^2=1$.

1 Check Assertion: write each $\sin^2$ as $1-\cos^2$.
$\sin^2\alpha+\sin^2\beta+\sin^2\gamma=3-(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)$

2	Use $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$. So A is true. $=3-1=2$	
3	Check Reason: $l^2+m^2+n^2=1$ is a standard result. So R is true.	
4	R is exactly the fact used to prove A, so R explains A.	✓ 1
5	Hence the answer is option (A).	

Answer **(A) Both true; R is the correct explanation of A.**

Watch out: Choosing (B): R is not a separate fact here, it is the very step that gives $3-1=2$.

7 **Short answer** PYQ-style 2 marks

Find the vector and Cartesian equations of the line passing through the points $(3,4,-7)$ and $(1,-1,6)$.

Formula / idea: Line through $\vec{a}$ and $\vec{b}$: $\vec{r}=\vec{a}+\lambda(\vec{b}-\vec{a})$.

1	Write the position vectors of the two points. $\vec{a}=3\hat{i}+4\hat{j}-7\hat{k},\quad \vec{b}=\hat{i}-\hat{j}+6\hat{k}$	✓ ½
2	Find the direction vector $\vec{b}-\vec{a}$. $\vec{b}-\vec{a}=-2\hat{i}-5\hat{j}+13\hat{k}$	✓ ½
3	Write the vector equation. $\vec{r}=(3\hat{i}+4\hat{j}-7\hat{k})+\lambda(-2\hat{i}-5\hat{j}+13\hat{k})$	✓ ½
4	Write the Cartesian equation using point $(3,4,-7)$ and DRs $-2,-5,13$. $\frac{x-3}{-2}=\frac{y-4}{-5}=\frac{z+7}{13}$	✓ ½

Answer $\vec{r}=(3\hat{i}+4\hat{j}-7\hat{k})+\lambda(-2\hat{i}-5\hat{j}+13\hat{k})$; $\frac{x-3}{-2}=\frac{y-4}{-5}=\frac{z+7}{13}$

Watch out: Writing $\vec{r}=\vec{a}+\lambda\vec{b}$, using the second point as the direction instead of $\vec{b}-\vec{a}$.

Level 2

16 marks

8 **Short answer** PYQ-style 2 marks

Show that the line through the points $(1,-1,2)$ and $(3,4,-2)$ is perpendicular to the line through the points $(0,3,2)$ and $(3,5,6)$.

Formula / idea: DRs of the line through two points are $x_2-x_1,\ y_2-y_1,\ z_2-z_1$.

1	DRs of the first line. $3-1,\ 4-(-1),\ -2-2=2,\ 5,\ -4$	✓ ½

2	DRs of the second line. $3-0,\ 5-3,\ 6-2 = 3,\ 2,\ 4$	✓ ½
3	Find $a_1a_2 + b_1b_2 + c_1c_2$. $2(3) + 5(2) + (-4)(4) = 6 + 10 - 16 = 0$	✓ ½
4	Hence the two lines are perpendicular.	✓ ½

Answer $a_1a_2 + b_1b_2 + c_1c_2 = 6 + 10 - 16 = 0$**, so the lines are perpendicular.**

Watch out: Using the coordinates of one point as DRs instead of the differences of the two points.

9 **Short answer** CBSE SQP 2024-25 3 marks

Find the shortest distance between the lines l_1 and l_2 whose vector equations are $\vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \lambda(7\hat{i} - 6\hat{j} + \hat{k})$ and $\vec{r} = (3\hat{i} + 5\hat{j} + 7\hat{k}) + \mu(\hat{i} - 2\hat{j} + \hat{k})$, where λ and μ are parameters.

Formula / idea: For skew lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$, $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$: $d = \left|\dfrac{(\vec{b}_1 \times \vec{b}_2)\cdot(\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}\right|$.

1	Compare with $\vec{r} = \vec{a} + \lambda\vec{b}$. $\vec{a}_1 = -\hat{i} - \hat{j} - \hat{k}, \quad \vec{b}_1 = 7\hat{i} - 6\hat{j} + \hat{k}$ $\vec{a}_2 = 3\hat{i} + 5\hat{j} + 7\hat{k}, \quad \vec{b}_2 = \hat{i} - 2\hat{j} + \hat{k}$	
2	Find $\vec{a}_2 - \vec{a}_1$. $\vec{a}_2 - \vec{a}_1 = 4\hat{i} + 6\hat{j} + 8\hat{k}$	✓ ½
3	Find $\vec{b}_1 \times \vec{b}_2$. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix} = -4\hat{i} - 6\hat{j} - 8\hat{k}$	✓ 1
4	Find its magnitude. $\lvert\vec{b}_1 \times \vec{b}_2\rvert = \sqrt{16 + 36 + 64} = \sqrt{116} = 2\sqrt{29}$	✓ ½
5	Find the dot product with $\vec{a}_2 - \vec{a}_1$. $(-4)(4) + (-6)(6) + (-8)(8) = -116$	✓ ½
6	Put into the formula. $d = \left\lvert\dfrac{-116}{2\sqrt{29}}\right\rvert = \dfrac{58}{\sqrt{29}} = 2\sqrt{29}$ units	✓ ½

Answer $2\sqrt{29}$ **units**

Watch out: Keeping the minus sign: distance is the modulus, so $-2\sqrt{29}$ must become $2\sqrt{29}$.

10 **Short answer** CBSE 2024 — 3 marks

Find the distance between the line $\frac{x}{2} = \frac{2y-6}{4} = \frac{1-z}{-1}$ and another line parallel to it passing through the point $(4, 0, -5)$.

Formula / idea: Parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$, $\vec{r} = \vec{a}_2 + \mu\vec{b}$: $d = \dfrac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$.

1 Write the given line in standard form. ✓ ½
$\frac{x-0}{2} = \frac{y-3}{2} = \frac{z-1}{1}$

2 Read the point, direction and the second point.
$\vec{a}_1 = 3\hat{j} + \hat{k}, \quad \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}, \quad \vec{a}_2 = 4\hat{i} - 5\hat{k}$

3 Find $\vec{a}_2 - \vec{a}_1$. ✓ ½
$\vec{a}_2 - \vec{a}_1 = 4\hat{i} - 3\hat{j} - 6\hat{k}$

4 Find $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$. ✓ 1
$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 4 & -3 & -6 \end{vmatrix} = -9\hat{i} + 16\hat{j} - 14\hat{k}$

5 Find the two magnitudes. ✓ ½
$\sqrt{81 + 256 + 196} = \sqrt{533}, \qquad |\vec{b}| = \sqrt{4 + 4 + 1} = 3$

6 Put into the formula. ✓ ½
$d = \frac{\sqrt{533}}{3}$ units

Answer $\frac{\sqrt{533}}{3}$ **units**

Watch out: Taking DRs $2, 4, -1$ straight from the given form; make the coefficients of y and z equal to 1 first.

11 **Short answer** PYQ-style — 2 marks

Find the angle between the lines $\vec{r} = 3\hat{i} + \hat{j} - 2\hat{k} + \lambda(\hat{i} - \hat{j} - 2\hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - 56\hat{k} + \mu(3\hat{i} - 5\hat{j} - 4\hat{k})$.

Formula / idea: Angle between lines: $\cos\theta = \left|\dfrac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}\right|$.

1 Write the direction vectors. ✓ ½
$\vec{b}_1 = \hat{i} - \hat{j} - 2\hat{k}, \quad \vec{b}_2 = 3\hat{i} - 5\hat{j} - 4\hat{k}$

2 Find the dot product and the magnitudes. ✓ ½
$\vec{b}_1 \cdot \vec{b}_2 = 3 + 5 + 8 = 16, \quad |\vec{b}_1| = \sqrt{6}, \quad |\vec{b}_2| = \sqrt{50} = 5\sqrt{2}$

3 Put into the formula. ✓ ½

$$\cos\theta = \frac{16}{\sqrt{6}\cdot 5\sqrt{2}} = \frac{16}{10\sqrt{3}} = \frac{8\sqrt{3}}{15}$$

4 Write the angle. ✓ ½

$$\theta = \cos^{-1}\left(\frac{8\sqrt{3}}{15}\right)$$

Answer $\theta = \cos^{-1}\left(\frac{8\sqrt{3}}{15}\right)$

Watch out: Using the position vectors $3\hat{i}+\hat{j}-2\hat{k}$ and $2\hat{i}-\hat{j}-56\hat{k}$ instead of the direction vectors.

12 **Short answer** PYQ-style 2 marks

Find the value of k so that the lines $x = -y = kz$ and $x-2 = 2y+1 = -z+1$ are perpendicular to each other.

Formula / idea: Perpendicular lines: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

1 Write the first line in standard form. ✓ ½

$$\frac{x}{1} = \frac{y}{-1} = \frac{z}{\frac{1}{k}} \Rightarrow \text{DRs } 1, -1, \frac{1}{k}$$

2 Write the second line in standard form. ✓ ½

$$\frac{x-2}{1} = \frac{y+\frac{1}{2}}{\frac{1}{2}} = \frac{z-1}{-1} \Rightarrow \text{DRs } 1, \frac{1}{2}, -1$$

3 Apply the perpendicular condition. ✓ ½

$$1(1) + (-1)\left(\frac{1}{2}\right) + \frac{1}{k}(-1) = 0$$

4 Solve for k. ✓ ½

$$\frac{1}{2} - \frac{1}{k} = 0 \Rightarrow k = 2$$

Answer $k = 2$

Watch out: Taking DRs $1, -1, k$ from $x = -y = kz$; the DR for z is $\frac{1}{k}$, not k.

13 **Short answer** CBSE 2023 2 marks

If the equation of a line is $x = ay + b$, $z = cy + d$, then find the direction ratios of the line and a point on the line.

Formula / idea: Bring the line to $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$; then read the DRs and the point.

1	Make y the subject of each equation. $x = ay + b \Rightarrow y = \frac{x - b}{a}, \qquad z = cy + d \Rightarrow y = \frac{z - d}{c}$	
2	Equate the three expressions for y. $\frac{x - b}{a} = \frac{y - 0}{1} = \frac{z - d}{c}$	✓ 1
3	Read the direction ratios from the denominators. $\text{DRs} = a,\ 1,\ c$	✓ ½
4	Read a point from the numerators. $(b,\ 0,\ d)$	✓ ½

Answer **DRs $a, 1, c$; a point on the line is $(b, 0, d)$.**

Watch out: Writing the DRs as a, b, c; the y-term has denominator 1 and b is part of the point.

14 **Short answer** CBSE SQP 2026-27 2 marks

A line passing through the points $(2, 6, 5)$ and $(3, p, 7)$ is parallel to the line joining the points $(-4, 3, 4)$ and $(-1, 0, q)$. Find the value of $\frac{q}{p}$.

Formula / idea: Parallel lines have proportional direction ratios: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

1	DRs of the first line. $3 - 2,\ p - 6,\ 7 - 5 = 1,\ p - 6,\ 2$	✓ ½
2	DRs of the second line. $-1 + 4,\ 0 - 3,\ q - 4 = 3,\ -3,\ q - 4$	✓ ½
3	Parallel lines have proportional DRs. $\frac{1}{3} = \frac{p - 6}{-3} = \frac{2}{q - 4}$	✓ ½

4 Solve each pair and find $\frac{q}{p}$. ✓ ½

$p-6=-1 \Rightarrow p=5, \quad q-4=6 \Rightarrow q=10, \quad \frac{q}{p}=2$

Answer $\frac{q}{p}=2$

Watch out: Using the perpendicular condition $a_1a_2+b_1b_2+c_1c_2=0$ for parallel lines.

Level 3

27 marks

15 **Long answer** CBSE SQP 2026-27 5 marks

Find the foot of the perpendicular drawn from the point $(2,1,4)$ on the line $x-2=\frac{2y+3}{4}=2-z$. Also find the equation of the line passing through this point and the point $(2,1,4)$.

Formula / idea: The foot P lies on the line and $\overrightarrow{AP}$ is perpendicular to the line's direction.

1 Write the line in standard form. ✓ ½

$\frac{x-2}{1}=\frac{y+\frac{3}{2}}{2}=\frac{z-2}{-1} \Rightarrow \text{DRs } 1,2,-1$

2 Put each fraction equal to k to get a general point. ✓ 1

$P\left(k+2,\ 2k-\frac{3}{2},\ 2-k\right)$

3 DRs of AP, where $A(2,1,4)$. ✓ ½

$k,\ 2k-\frac{5}{2},\ -k-2$

4 AP is perpendicular to the line. ✓ ½

$1(k)+2\left(2k-\frac{5}{2}\right)+(-1)(-k-2)=0$

5 Solve for k. ✓ ½

$k+4k-5+k+2=0 \ \Rightarrow\ 6k=3 \ \Rightarrow\ k=\frac{1}{2}$

6 Put $k=\frac{1}{2}$ in P to get the foot. ✓ 1

$P\left(\frac{5}{2},\ -\frac{1}{2},\ \frac{3}{2}\right)$

7 DRs of AP at $k=\frac{1}{2}$; multiply by 2. ✓ ½

$\frac{1}{2},\ -\frac{3}{2},\ -\frac{5}{2} \ \Rightarrow\ 1,\ -3,\ -5$

8 Equation of the line through $A(2,1,4)$. ✓ ½

$$\frac{x-2}{1}=\frac{y-1}{-3}=\frac{z-4}{-5}$$

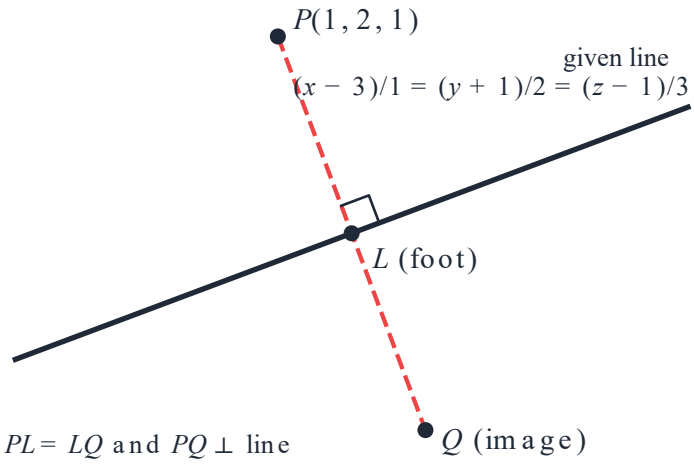
Answer **Foot** $P\left(\frac{5}{2},-\frac{1}{2},\frac{3}{2}\right)$; **line** $\frac{x-2}{1}=\frac{y-1}{-3}=\frac{z-4}{-5}$

Watch out: Taking DRs $1,4,1$ from $x-2=\frac{2y+3}{4}=2-z$ without rewriting; the correct DRs are $1,2,-1$.

16 **Long answer** CBSE SQP 2024-25 5 marks

Find the image of the point $(1,2,1)$ with respect to the line $\frac{x-3}{1}=\frac{y+1}{2}=\frac{z-1}{3}$. Also find the equation of the line joining the given point and its image.

Formula / idea: Foot L of the perpendicular from P is the mid-point of P and its image Q.



1 Let $P(1,2,1)$; put the line equal to λ for a general point L. ✓ 1

$$L(\lambda+3,\ 2\lambda-1,\ 3\lambda+1)$$

2 DRs of PL. ✓ ½

$$\lambda+2,\ 2\lambda-3,\ 3\lambda$$

3 PL is perpendicular to the line (DRs $1,2,3$).

$$1(\lambda+2)+2(2\lambda-3)+3(3\lambda)=0$$

4 Solve for λ. ✓ 1

$$14\lambda-4=0\ \Rightarrow\ \lambda=\frac{2}{7}$$

5 Foot of the perpendicular. ✓ ½

$$L\left(\frac{23}{7},-\frac{3}{7},\frac{13}{7}\right)$$

6 L is the mid-point of PQ, with $Q(x_1, y_1, z_1)$.
$\frac{1+x_1}{2} = \frac{23}{7}, \quad \frac{2+y_1}{2} = -\frac{3}{7}, \quad \frac{1+z_1}{2} = \frac{13}{7}$

7 Solve to get the image. ✓ 1
$Q\left(\frac{39}{7}, -\frac{20}{7}, \frac{19}{7}\right)$

8 DRs of PQ; multiply by $\frac{7}{2}$.
$\frac{32}{7}, -\frac{34}{7}, \frac{12}{7} \Rightarrow 16, -17, 6$

9 Equation of the line through $P(1, 2, 1)$. ✓ 1
$\frac{x-1}{16} = \frac{y-2}{-17} = \frac{z-1}{6}$

Answer **Image** $Q\left(\frac{39}{7}, -\frac{20}{7}, \frac{19}{7}\right)$**; line** PQ**:** $\frac{x-1}{16} = \frac{y-2}{-17} = \frac{z-1}{6}$

Watch out: Stopping at the foot L and calling it the image; the image is $Q = 2L - P$.

17 **Long answer** CBSE 2025 5 marks

Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.

Formula / idea: Lines meet if their general points agree for some λ, μ in all three coordinates.

1 General point on the first line. ✓ ½
$(2\lambda + 1,\ 3\lambda + 2,\ 4\lambda + 3)$

2 General point on the second line. ✓ ½
$(5\mu + 4,\ 2\mu + 1,\ \mu)$

3 Equate the x and y coordinates. ✓ ½
$2\lambda - 5\mu = 3, \qquad 3\lambda - 2\mu = -1$

4 Solve the two equations. ✓ 1
$\lambda = -1, \quad \mu = -1$

5 Check the z coordinates. ✓ ½
$4(-1) + 3 = -1 = \mu$ ✓
All three agree, so the lines intersect.

6 Find the point of intersection. ✓ ½
$(2(-1) + 1,\ 3(-1) + 2,\ 4(-1) + 3) = (-1, -1, -1)$

7 Use the distance formula with $(-1, -5, -10)$. ✓ 1
$\sqrt{(-1+1)^2 + (-1+5)^2 + (-1+10)^2} = \sqrt{0 + 16 + 81}$

8 Hence the required distance.
$= \sqrt{97}$ units ✓ ½

Answer **Point of intersection** $(-1, -1, -1)$**; distance** $= \sqrt{97}$ **units**

Watch out: Skipping the z-check; without it you have not shown that the lines actually meet.

18 **Long answer** CBSE SQP 2025-26 5 marks

The two lines $\frac{x-1}{3} = -y,\ z+1=0$ and $\frac{-x}{2} = \frac{y+1}{2} = z+2$ intersect at a point whose y-coordinate is 1. Find the coordinates of their point of intersection. Find the vector equation of a line perpendicular to both the given lines and passing through this point of intersection.

Formula / idea: A line perpendicular to both $\vec{b}_1$ and $\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$.

1 General point on the first line: put $\frac{x-1}{3} = -y = t$, with $z = -1$.
$(3t+1,\ -t,\ -1)$ ✓ 1

2 Its y-coordinate is 1.
$-t = 1 \Rightarrow t = -1 \Rightarrow (-2,\ 1,\ -1)$ ✓ ½

3 Check this point on the second line.
$\frac{-(-2)}{2} = 1, \quad \frac{1+1}{2} = 1, \quad -1+2 = 1$
All three are equal, so the point lies on both lines. ✓ 1

4 Direction vectors of the two lines.
$\vec{b}_1 = 3\hat{i} - \hat{j} + 0\hat{k}, \quad \vec{b}_2 = -2\hat{i} + 2\hat{j} + \hat{k}$
Second line: $\frac{x}{-2} = \frac{y+1}{2} = \frac{z+2}{1}$. ✓ ½

5 Find $\vec{b}_1 \times \vec{b}_2$.
$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 0 \\ -2 & 2 & 1 \end{vmatrix} = -\hat{i} - 3\hat{j} + 4\hat{k}$ ✓ 1

6 Take the direction as $\hat{i} + 3\hat{j} - 4\hat{k}$ and write the equation.
$\vec{r} = (-2\hat{i} + \hat{j} - \hat{k}) + \lambda(\hat{i} + 3\hat{j} - 4\hat{k})$ ✓ 1

Answer **Point** $(-2, 1, -1)$**;** $\vec{r} = (-2\hat{i} + \hat{j} - \hat{k}) + \lambda(\hat{i} + 3\hat{j} - 4\hat{k})$

Watch out: Reading the first line's DRs as $3, -1, 1$; $z + 1 = 0$ means z never changes, so its DR is 0.

19 **Short answer** CBSE SQP 2024-25 3 marks

Find the vector and the Cartesian equations of the line that passes through $(-1, 2, 7)$ and is perpendicular to the lines $\vec{r} = 2\hat{i} + \hat{j} - 3\hat{k} + \lambda(\hat{i} + 2\hat{j} + 5\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 7\hat{k} + \mu(3\hat{i} - 2\hat{j} + 5\hat{k})$.

Formula / idea: A line perpendicular to both $\vec{b}_1$ and $\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$.

1 Write the two direction vectors. ✓ ½

$\vec{b}_1 = \hat{i} + 2\hat{j} + 5\hat{k}, \quad \vec{b}_2 = 3\hat{i} - 2\hat{j} + 5\hat{k}$

2 Find $\vec{b}_1 \times \vec{b}_2$. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 5 \\ 3 & -2 & 5 \end{vmatrix} = 20\hat{i} + 10\hat{j} - 8\hat{k} = 2(10\hat{i} + 5\hat{j} - 4\hat{k})$$

3 Vector equation through $(-1, 2, 7)$. ✓ 1

$\vec{r} = (-\hat{i} + 2\hat{j} + 7\hat{k}) + \lambda(10\hat{i} + 5\hat{j} - 4\hat{k})$

4 Cartesian equation. ✓ ½

$$\frac{x+1}{10} = \frac{y-2}{5} = \frac{z-7}{-4}$$

Answer $\vec{r} = (-\hat{i} + 2\hat{j} + 7\hat{k}) + \lambda(10\hat{i} + 5\hat{j} - 4\hat{k})$; $\dfrac{x+1}{10} = \dfrac{y-2}{5} = \dfrac{z-7}{-4}$

Watch out: Sign slip in the middle term of the determinant: it is $-\hat{j}(5 - 15) = +10\hat{j}$.

20 **Case study** CBSE SQP 2026-27 4 marks

Case Study: Two security sensors in a multi-storey mall emit laser beams for movement detection. Three of these beams travel along the following lines: $l_1 : \vec{r} = (\lambda + 1)\hat{i} + (3\lambda + 2)\hat{j} + (3 - \lambda)\hat{k}$, $l_2 : \vec{r} = 2(\mu + 1)\hat{i} + (3\mu - 1)\hat{j} + (\mu + 1)\hat{k}$ and $l_3 : \vec{r} = (\gamma + 3)\hat{i} + (3\gamma - 1)\hat{j} + (1 - \gamma)\hat{k}$. Based on the given information, answer the following questions.

(i) Show that the lines l_1 and l_2 are not parallel to each other. 1 mark

(ii) Show that the lines l_1 and l_3 are parallel to each other. 1 mark

(iii) Find the shortest distance between the lines l_1 and l_2. 2 marks

Formula / idea: Separate the constant and parameter parts: $\vec{r} = \vec{a} + \lambda\vec{b}$; parallel lines have proportional $\vec{b}$.

(i) 1 mark

1 Write l_1 and l_2 in the form $\vec{a} + \lambda\vec{b}$. ✓ ½

$l_1 : (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} + 3\hat{j} - \hat{k})$

$l_2 : (2\hat{i} - \hat{j} + \hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$

2 Compare the ratios of the direction ratios. ✓ ½

$\dfrac{1}{2} \neq \dfrac{3}{3}$

3 DRs are not proportional, so l_1 and l_2 are not parallel.

Answer (i) $\frac{1}{2} \neq \frac{3}{3}$, **so** l_1 **and** l_2 **are not parallel.**

(ii)

1 mark

1 Write l_3 in the form $\vec{a} + \gamma\vec{b}$. ✓ ½

$l_3 : (3\hat{i} - \hat{j} + \hat{k}) + \gamma(\hat{i} + 3\hat{j} - \hat{k})$

2 Both l_1 and l_3 have direction $\hat{i} + 3\hat{j} - \hat{k}$, so they are parallel. ✓ ½

Answer (ii) **Same direction vector** $\hat{i} + 3\hat{j} - \hat{k}$**, so** $l_1 \parallel l_3$**.**

(iii)

2 marks

1 Find $\vec{a}_2 - \vec{a}_1$.

$(2\hat{i} - \hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = \hat{i} - 3\hat{j} - 2\hat{k}$

2 Find $\vec{b}_1 \times \vec{b}_2$ and its magnitude. ✓ 1

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -1 \\ 2 & 3 & 1 \end{vmatrix} = 6\hat{i} - 3\hat{j} - 3\hat{k}, \quad |\vec{b}_1 \times \vec{b}_2| = 3\sqrt{6}$$

3 Find the dot product. ✓ ½

$(6\hat{i} - 3\hat{j} - 3\hat{k}) \cdot (\hat{i} - 3\hat{j} - 2\hat{k}) = 6 + 9 + 6 = 21$

4 Put into the shortest distance formula. ✓ ½

$$d = \left| \frac{21}{3\sqrt{6}} \right| = \frac{7}{\sqrt{6}} = \frac{7\sqrt{6}}{6} \text{ units}$$

Answer (iii) $\frac{7\sqrt{6}}{6}$ **units**

Watch out: Reading l_2's direction as $\hat{i} + 3\hat{j} + \hat{k}$: in $2(\mu + 1)\hat{i}$ the μ-coefficient is 2.